

Analysis of Ramming Process for 155 mm×52 Caliber Gun Barrel System

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ABSTRACT

The interaction between rotating band and barrel during the ramming and engraving process influences the performance of gun. Several studies for the latter process have been reported, while studies for the former are limited to a few gun systems, including 152 mm mod 77, 125 mm T-72 tank, 125 mm 2A46 tank, and have not undertaken for 155 mm×52 caliber gun. Therefore, finite element simulations of ramming process have been performed for new and worn-out barrel of 155 mm×52 caliber gun system. The effect of leading angle of band on the contact force is also studied. Results show that ramming velocity V_r of 5 m/s generates enough contact force to avoid the fallback of projectile for maximum angle of barrel-elevation. Also, contact force drops gradually as barrel wear increases till 0.5 %, but drops rapidly for further enhancement in barrel wear. The minimum V_r ensuring no fallback in a worn barrel increases marginally up to 1 % barrel-wear, but it increases exponentially for further rise in barrel wear. Further, the driving band with a leading angle of 3° is found to produce maximum contact pressure. The results obtained from this study would help in understanding the ramming process better for the 155mm × 52 mm Caliber gun.

Keywords: Rotating band; Barrel wear; Ramming velocity; Contact force

NOMENCLATURE

u_θ	: Tangential displacement
V_r	: Initial ramming velocity
A	: Yield stress at reference conditions
B	: Strain hardening modulus
C	: Strain rate sensitivity coefficient
n	: Strain hardening exponent
m	: Thermal softening coefficient
D_1	: Failure strain parameter
D_2	: Exponential factor scaling influence of stress triaxiality
D_3	: Triaxiality sensitivity factor
D_4	: Strain rate sensitivity for fracture
D_5	: Temperature sensitivity for fracture
β	: Inclination of the gun barrel with respect to the ground level
V_z	: Projectile velocity inside barrel
F_c	: Contact force between barrel and band
d_i	: Inner land diameter of barrel
α	: Inclination angle of the leading surface of the band (leading angle)
ϵ_p	: Equivalent plastic strain

1. INTRODUCTION

The rotating band is used to ensure proper positioning of

the projectile, obturation against burned gases, and to impart spin motion to the projectile¹⁻³. The contact force (or retention force) between band and barrel-forcing cone developed after ramming must be strong enough to avoid the fallback of the projectile for the highest elevation angles^{1,4}. The repeated firing causes continuous barrel wear^{1,5-6} which may lead to improper projectile seating and a drop in retention force³ leading to poor performance of the gun system. Therefore, a thorough understanding of the band deformation during ramming is essential.

Various experimental⁷⁻¹¹ and numerical studies on the engraving process of different projectiles for new barrel¹²⁻²² and worn barrel^{5,19} have been undertaken. By contrast, only a few studies have focused on analyzing the ramming process¹⁻⁴ till now. Balla, *et al.*⁴ conducted finite element (FE) simulations of the ramming process of a high-explosive projectile and a shaped charge penetration projectile in a self-propelled howitzer (152 mm mod 77) and 125 mm T-72 tank cannon, respectively, to analyze the plastic deformation of the driving band and to determine the reaction force between barrel and band during the ramming process. The ramming process of Armour-Piercing Fin-Stabilized Discarding Sabot (APFSDS) projectiles for new as well as worn 125 mm smoothbore tank cannon barrels have also been analyzed by Balla¹, *et al.* through FE simulations considering thermal effects. They showed that barrel wear has a more pronounced influence on the ramming process than thermal deformations. In addition, they observed that the distance travelled by the projectile during ramming

was significantly greater in a worn barrel than that of a new barrel, resulting in an increased chamber volume and changes in burned gas pressure. Further, Balla³, *et al.* also performed experimental investigations on the 125 mm 2A46 tank cannon to determine the force required to extract the projectile from the barrel to the forcing cone, after the ramming process.

It is important to notice from the above discussion that the analysis of the ramming process has been performed only for a few gun systems it has not been undertaken for the 155 mm $\times$ 52 mm Caliber gun system. Therefore, though some understanding of the deformation behaviour of the rotating band during the ramming process has been gained for other guns, the outcome of these studies may not be directly applicable to the 155 mm $\times$ 52 mm Caliber gun system. Also, there are several fundamental questions whose answer would help in designing a better rotating band for the 155 mm $\times$ 52 mm Caliber gun system. For example, how does the retention force in the 155 mm $\times$ 52 mm Caliber gun system evolve during the ramming process? How does it change with a change in the leading angle of the rotating band? Does the retention force drop linearly with the increase in barrel wear? What should be the initial ramming velocity for a worn-out barrel to avoid the fallback of ammunition, and how does it vary with barrel wear?

In order to address these issues, FE simulations of the ramming process are performed on a 155 mm $\times$ 52 mm Caliber gun system. The performance of FE simulations is benchmarked by comparing simulated engraving marks with experimental data. Results show a nonlinear dependence of retention force and initial ramming velocity on barrel wear. The outcome of this study would help understand the ramming process for the 155 mm $\times$ 52 mm Caliber gun system.

2. FINITE ELEMENT MODEL

The 3D models of gun barrel, shell and rotating band are generated by employing parametric modelling, presented by Ding⁵, *et al.*, in software Abaqus 2017. Since the projectile travels a very short distance inside the forcing cone during the ramming operation, the influence of helix angle of barrel rifling on the band deformation during ramming is expected to be insignificant, therefore, helix angle is taken to be zero in this

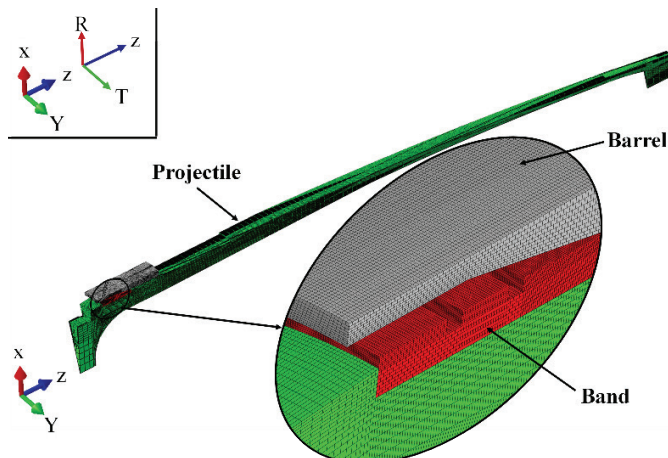


Figure 1. The finite element model of the assembly 15° sector-section of shell, rotating band and barrel employed in the analysis of the ramming process.

study. Further, it is also assumed that the cross-section of the barrel remains symmetrical about the axis of the barrel even after wearing of the barrel. Thus, considering the above stated assumptions, only a 15° sector-section of the shell, rotating band and barrel corresponding to two grooves and two lands are modelled. Further, to reduce computational time, instead of modelling the entire length of the barrel only the forcing cone of the barrel is modelled. The FE model of assembly of barrel, shell and rotating band employed in the analysis is depicted in Fig. 1.

The 3D 8-node linear brick elements with reduced integration (C3D8R) are used to discretise the rotating band, shell, and barrel. To capture the strain and stress gradient in the rotating band, a highly refined mesh with the smallest element size of 0.3 mm is employed to mesh the contacting surfaces of the rotating band and barrel rifling, refer to Fig. S1(a) and (b) in the supplemental material (SM). Also, the region in the projectile which is in contact with the band is finely meshed, while the remaining portion of the barrel and the projectile is discretized by coarser elements to reduce computational time.

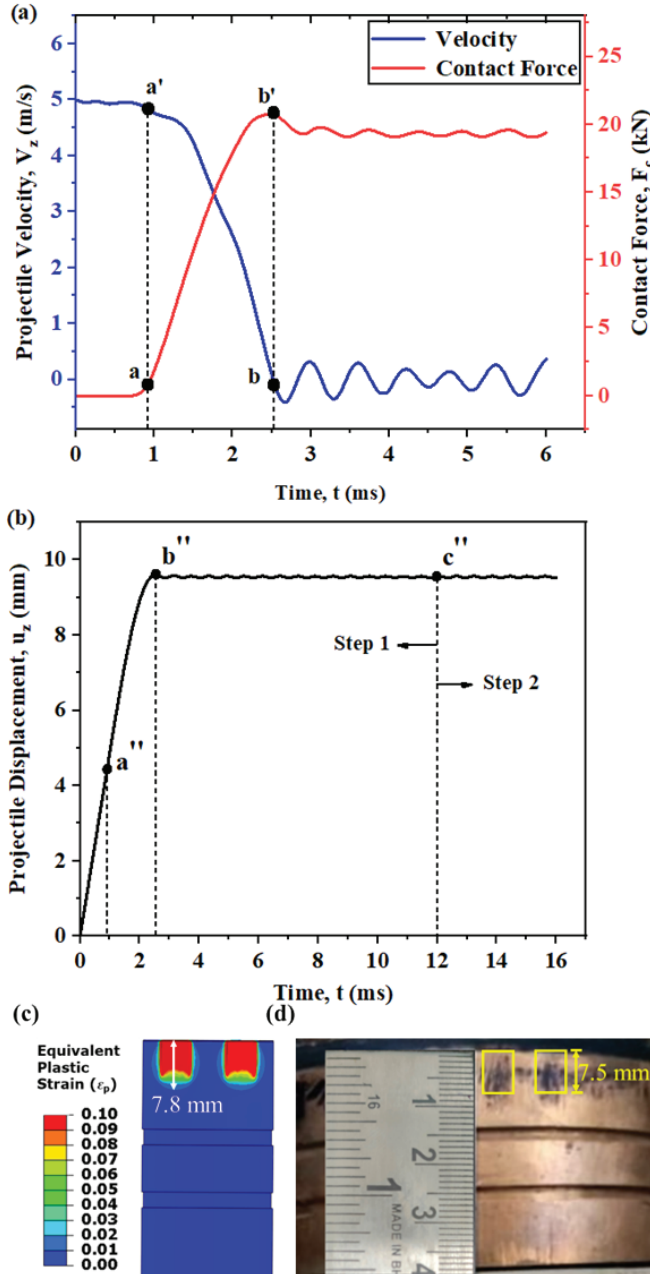
To enhance the stability and accuracy of calculations, the distortion control and stiffness hourglass control are enabled. A general contact (Explicit) algorithm in-built with Abaqus is used to model interaction between barrel and band as well as band and projectile. Further, normal and tangential behaviour are employed by assuming a coefficient of friction of 0.1 between meeting surfaces. The internal element-surfaces of the band are also defined to ensure contact is established between the barrel and the newly created surface on the band after deletion of the failed element in the band. In addition, a tie constraint is applied to restrict the slipping between the projectile and the band during deformation.

All the nodes on the diametral face of the barrel on the breach end are constrained to move along X, Y and Z directions. Further, noting the symmetry in deformation of the shell, band and barrel about their side faces, the tangential displacement, $u_\theta = 0$, is prescribed for all the nodes lying on these faces. Here, u_θ is defined with respect to a local coordinate system (R, θ , Z) as shown in Fig. 1. Note that the Z axes of the local and global coordinate systems are parallel. The body forces corresponding to the gravitational field are applied on the shell and band to incorporate the effect of barrel inclination. In all simulations, β is taken to be 72°. Also, simulations are performed in two steps. In the first step V_s along the Z direction is assigned to all the nodes in the shell. The duration of this step is kept sufficiently long to ensure that the initially moving projectile is positioned properly inside the barrel. In the second step, a pressure equivalent to the five times the weight of the projectile is applied on the nose of the projectile to analyse if the contact force generated between the band and barrel is sufficient to prevent the fallback of the projectile⁴. The force corresponding to the applied pressure in Step 2 is referred to as the extraction force in the following.

The shell and barrel are assumed to behave as a linear elastic material, whereas band material is assumed to follow the Johnson-Cook plasticity model²³, which is briefly described in SM. The material parameters used in simulations are listed in Table 1.

Table 1. Material parameters for rotating band taken from Sun²⁴, *et al.* (2015)

Parameter	Value	Parameter	Value
A	90 MPa	D ₁	0.54
B	292 MPa	D ₂	4.89
C	0.025	D ₃	3.03
n	0.31	D ₄	0.014
m	1.09	D ₅	1.12

**Figure 2.** (a) The evolution of projectile velocity, V_z and contact force, F_c between rotating band and barrel, (b) The evolution of the axial displacement of projectile, (c) The contour plots of equivalent plastic strain (ϵ_p), on the rotating band obtained from finite element simulations, and (d) The marks of deformation on an actual band of a 155 mm×52 caliber gun system.

3. RESULTS AND DISCUSSION

3.1 Analysis of Ramming of Projectile on a New Barrel

Analysis of the ramming process of a projectile rammed with $V_r = 5$ m/s on a new barrel is performed. Fig. 2(a) shows the evolution of V_z and F_c during ramming. Note that V_z remains almost identical to V_r up to time, $t = 0.93$ ms (point a), but begins to drop rapidly thereafter. F_c begins to increase at around the same time, suggesting the establishment of contact between band and barrel occurring at $t = 0.93$ ms (point a'). Due to a reduction in the diameter of the forcing cone, F_c continues to increase and attains its maximum at $t \sim 2.53$ ms (point b'). Consequently, V_z continues to drop and becomes zero at $t = 2.53$ ms (point b). Thus, Fig. 2(a) suggests that the ramming process gets over in around 2.53 ms in 155 mm × 52 caliber gun system. Note that oscillations in V_z and F_c after points b and b' are due to the fact that the impact of the band on the inner surface of the barrel and their further interaction is a non-linear dynamic process which is identified as the ping-pong effect²⁴.

Figure 2(b) shows projectile displacement inside the barrel during ramming. Here, points a'' and b'' correspond to the onset of contact and completion of ramming, respectively. Fig. 2(b) shows that the projectile has travelled a distance of ~ 4.5 mm inside the barrel before contact between the band and the barrel is established, while the total projectile displacement (or maximum projectile displacement) at the end of ramming is ~ 9 mm. There is no significant change in projectile displacement after ramming is over confirming that the projectile is successfully positioned inside the barrel. Further, point c'' in Fig. 2(b) corresponds to the start of simulations-step 2. Since there is no change in the projectile displacement even after the application of the extraction load, it can be concluded that the frictional contact force induced between band and barrel due to the ramming of the projectile with $V_r = 5$ m/s is sufficient enough to prevent the fallback of the projectile inside a barrel elevated at an angle of 72° . Figure 2(c) shows the contour plots of equivalent plastic strain at $t = 16$ ms (at the end of step 2), while Fig. 2(d) displays the marks of engraving on the band of a real projectile at the end of the ramming process. Fig. 2(c) clearly reveals a significant plastic deformation on the band at the end of ramming. Note, the shape and size of the deformed region of the band predicted by simulations are almost similar to those obtained from experiments. In particular, the length of the grooves on the engraved band is ~ 7.8 mm, which is in corroboration with the experimental result shown in Fig. 2(d).

3.2 Analysis of Ramming of Projectile on a Worn Barrel

In order to understand the effect of barrel wear on F_c , FE simulations of ramming are performed by increasing d_i . Assuming a linear relationship between the barrel wear and increase in d_i , simulations are performed by considering four values of d_i pertaining to 0.5 %-, 1 %- and 1.3 %- and 1.5 %-barrel wear. Fig. 3(a) shows the variation in F_c in four barrels pertaining to different wear conditions. It can be seen from Fig. 3(a) that F_c starts increasing at ~ 0.93 ms for a new barrel, while it begins at ~ 2.53 ms and ~ 6 ms for 0.5 % and

1.5 % wear of the barrel, respectively. Thus, it can be concluded the establishment of contact between the band and barrel gets delayed with an increase in barrel wear.

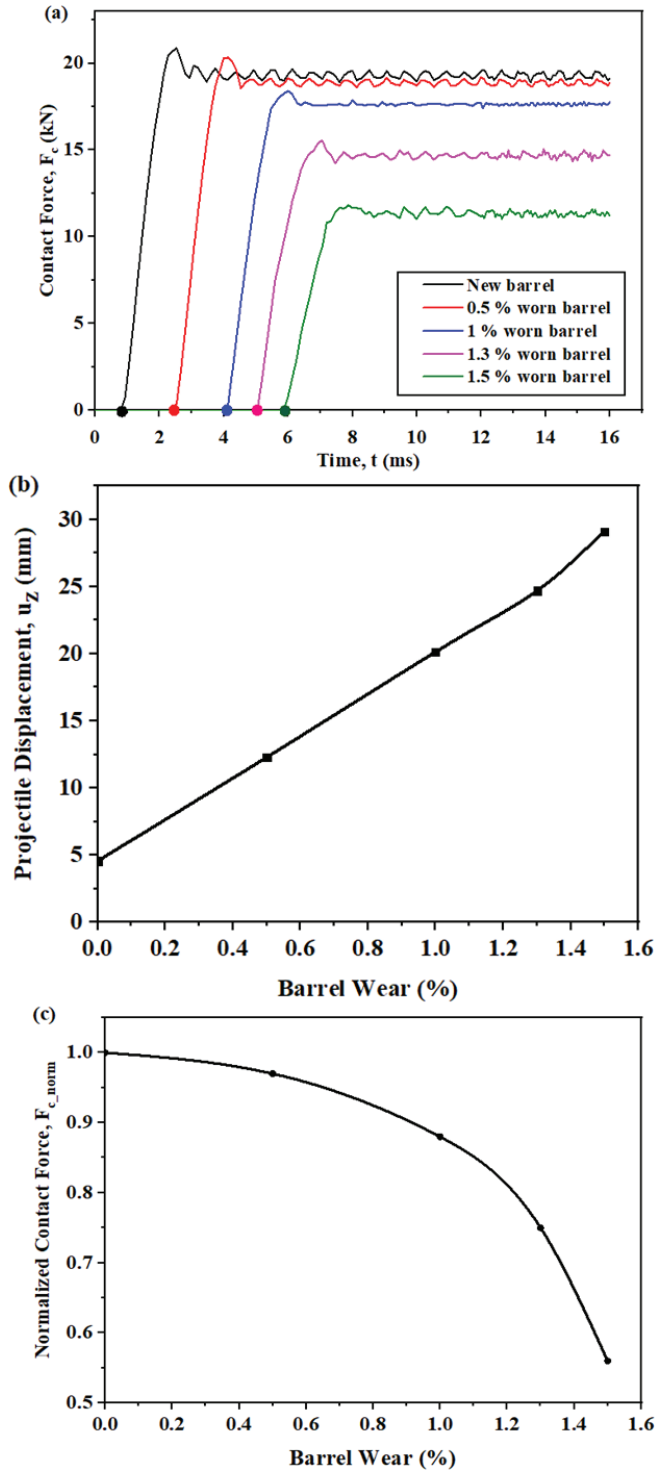


Figure 3. (a) The evolution of contact force in a new and worn barrel under identical ramming conditions, (b) The variation of displacement of projectile at the time of initiation of contact between band and barrel versus % barrel wear, and (c) The variation of normalized contact force, F_c/F_o with respect to % barrel wear. Here, F_o corresponds to the contact force in a new barrel.

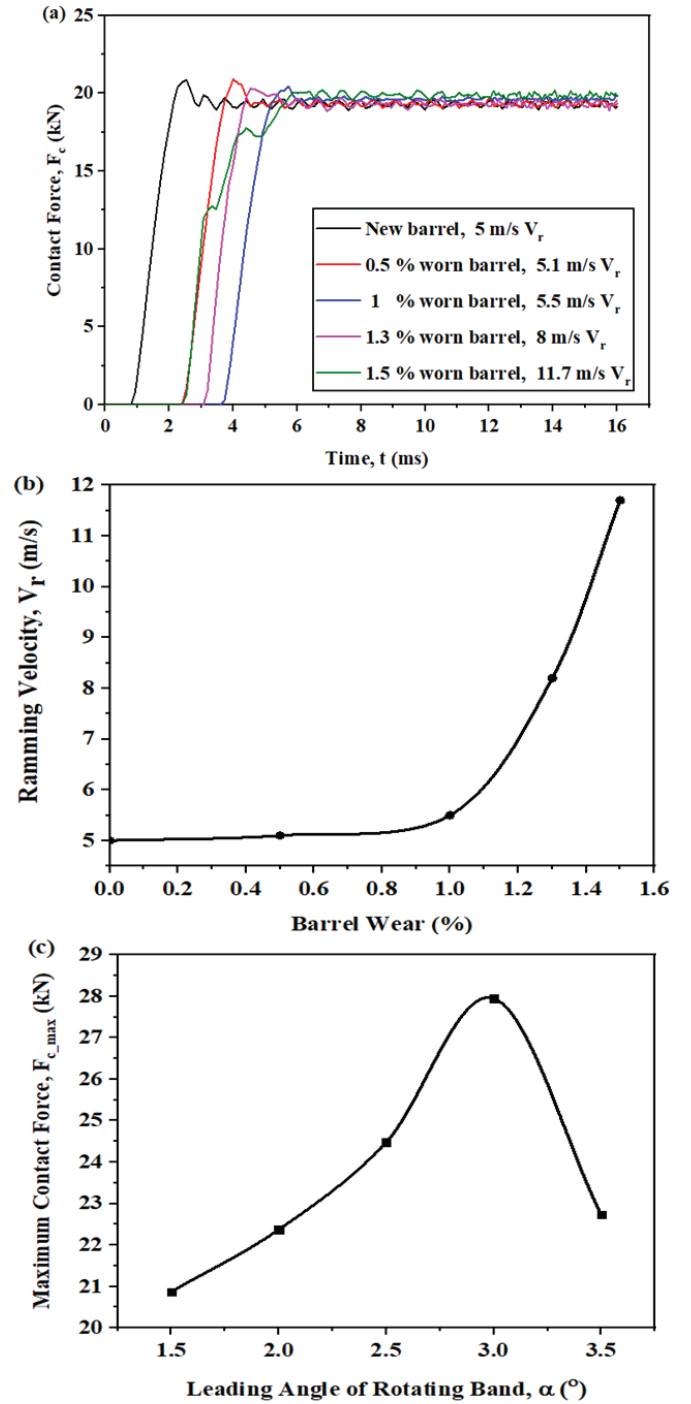


Figure 4. (a) The evolution of contact force corresponding to an initial ramming velocity which results in peak force in a worn barrel identical to that in a new barrel, (b) The variation of such required initial ramming velocity with barrel wear, and (c) The variation of maximum contact force with the leading angle of the rotating band.

The values of u_z at the onset of contact between band and barrel are plotted against percentage wear in barrel, in Fig. 3(b). Note that the distance travelled by the projectile before the contact is established increases almost linearly with an increase in barrel wear. The maximum contact force between the band and barrels and the percentage barrel wear are listed

in Table 2. To get more insights, the normalized contact force given as follows.

$$F_{c_norm} = F_{c_max} / F_o \quad (1)$$

where F_o is the maximum contact force in a new barrel, is computed and plotted against barrel wear in Fig. 3(c). Note, drop in F_{c_norm} is marginal up to 0.5% barrel wear, whereas it drops rapidly for further enhancement in barrel wear. Thus, it can be deduced from Fig. 3(c) that contact force drops nonlinearly with an increase in barrel wear, it drops gradually with an increase in barrel wear up to 0.5 %, whereas it reduces rapidly for further increase in barrel wear. If the barrel wear can be correlated with the number of shots fired from the barrel, then the life of the barrel associated with an acceptable drop in contact force can be estimated from Fig. 3(c).

A worn-out barrel must apply the same contact force as a new barrel to avoid the fall back of a projectile for the maximum angle of elevation. This can be achieved by enhancing the initial ramming velocity of the projectile. Therefore, it is advisable to determine a suitable ramming velocity for a worn barrel to achieve a contact force almost identical to that of a new barrel. To this end, many simulations of ramming operations are performed by varying V_r for a given percentage wear in a barrel. The value of V_r for which F_{c_max} in a worn barrel becomes identical to that for a new barrel is considered as a suitable ramming velocity for that worn barrel. Fig. 4(a) depicts the variation of F_c in worn barrels when ramming was performed with a suitable ramming velocity. The variation of these ramming velocities against percentage wear is shown in Fig. 4(b). It can be seen from this figure that the initial ramming velocity required to avoid the fallback of the projectile in a worn barrel increases marginally as the percentage wear increases up to 1 %. However, it increases exponentially for further increase in barrel wear.

3.3 The Effect of Leading Angle of Band on Contact Force

The geometry of the band considerably influences F_c . Since the effectiveness of ramming is primarily governed by F_c , the effect of band geometry on F_c needs to be analysed. One of the important parameters in band geometry governing this interaction is angle α . Therefore, in this section, the effect of α on F_c is studied. To this end, FE simulations of the ramming process are performed on bands with different values of α (refer to Fig. 1(c)) ranging from 1.5° to 3.5° . In all these simulations, the $V_r = 5$ m/s is taken. The peak value of F_c for these bands is plotted against α in Fig. 4(c), which shows that the contact force increases with α and attains a peak for $\alpha = 3^\circ$ and begins to drop thereafter.

To understand the mechanistic reasons for this trend, the distribution of ϵ_p on the upper surface of the rotating band is shown in Figs. 5(a)–(c) for $\alpha = 1.5^\circ$, 3° and 3.5° , respectively. It is interesting to notice that plastic deformation is mainly concentrated around the leading edge of the band for $\alpha = 1.5^\circ$ (see Fig. 5(a)), while it is concentrated near the first groove for $\alpha = 3.5^\circ$ (Fig. 5(c)). In contrast, the plastic deformation has spread over almost the entire region from the leading edge to the first groove of the band for $\alpha = 3^\circ$ (Fig. 5(b)). Thus, the

plastic zone size seems to be largest for $\alpha = 3^\circ$, followed by $\alpha = 3.5^\circ$ and 1.5° , though the values of plastic strain are higher for $\alpha = 3.5^\circ$. To estimate the contact area for different bands, the contour plots of contact pressure are shown in Fig. 5(d) – Fig. 5(f) for $\alpha = 1.5^\circ$, 3° , and 3.5° , respectively. The region on the surface of the band can be assumed not to be in contact with the barrel, where the contact pressure is zero. Thus, it can be noticed from Fig. 5(d) that the leading edge of the band is in contact with the barrel for $\alpha = 1.5^\circ$, while the region near

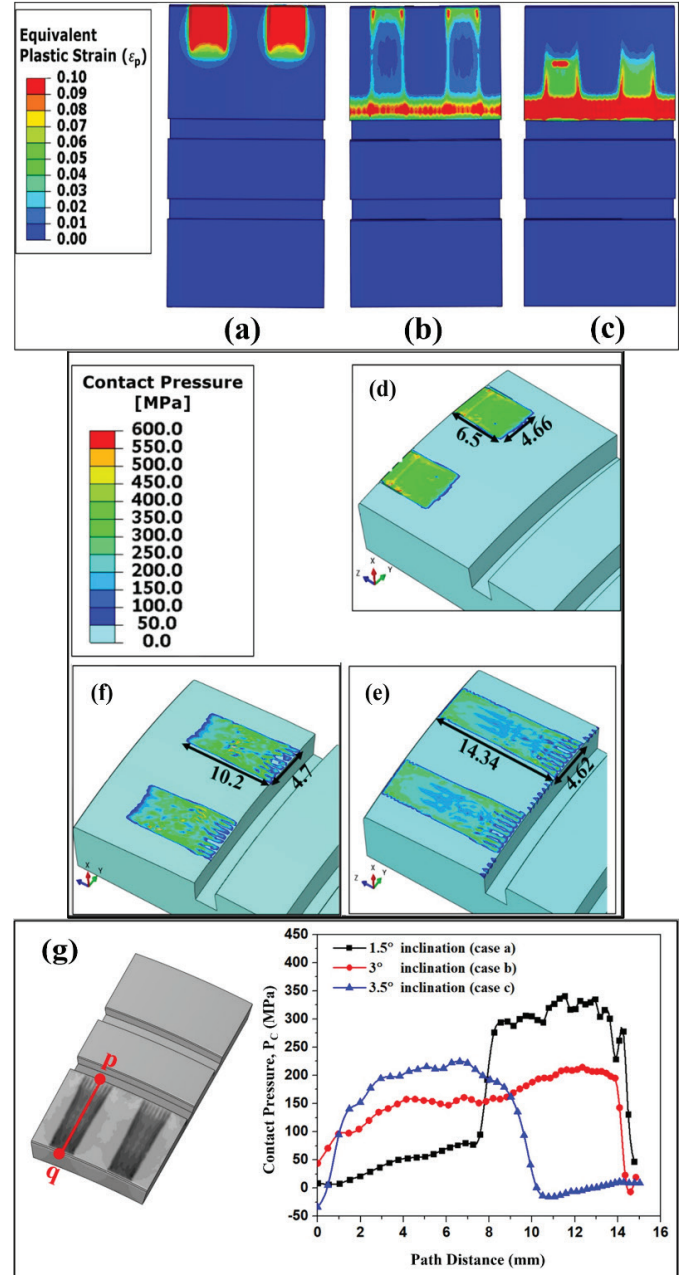


Figure 5. Contour plots of equivalent plastic strain, ϵ_p , at $t = 16$ ms on the top surface of the rotating band with leading angles (a) $\alpha = 1.5^\circ$, (b) $\alpha = 3^\circ$, and (c) $\alpha = 3.5^\circ$, respectively. Engraved profile on the leading surface of the bands with dimensions (in mm) for band of leading angles (d) 1.5° , (e) 3° , and (f) 3.5° respectively, and (g) The variation of contact pressure along the path pq on the top surface of the rotating band.

the first groove and not the leading edge is in contact with the barrel for $\alpha = 3.5^\circ$ (Fig. 5(f)). In contrast, almost the entire region from the leading edge to the first groove of the band contacts the barrel for $\alpha = 3^\circ$ (Fig. 5(e)). Further, the contact area for one engraving on the band surface is determined to be around 60.6 mm², 132.5 mm² and 95.9 mm² for $\alpha = 1.5^\circ$, 3° and 3.5° , respectively. It can also be noticed that though the contact pressure at local points for $\alpha = 3^\circ$ is not as high as for $\alpha = 1.5^\circ$ and 3.5° , the contact area is highest for the former than the latter cases, resulting in the highest contact force for $\alpha = 3^\circ$. This is in accordance with the variation of maximum contact force depicted in Fig. 4(c).

To get better contrast on the distribution of pressure for the three cases discussed above, contact pressure is extracted along a path 'p-q' and plotted against distance in Fig. 5(g). It can be noticed that the contact pressure for $\alpha = 1.5^\circ$ and 3.5° is higher than that for $\alpha = 3^\circ$ over some portion of the band, but it is almost zero for the remaining portion of the band. For $\alpha = 3^\circ$, pressure increases rapidly from point p, remains significantly large along the path, and then decreases as it approaches point q. In other words, contact pressure remains relatively uniform along path pq for $\alpha = 3^\circ$. As a result, the band experiences an evenly distributed pressure over the entire portion rather than extreme localized contact pressure. Thus, the $\alpha = 3^\circ$ is most preferable as there is no drastic deformation of the band at the same time, and the highest contact force is obtained at the band barrel interface. For the gun system analyzed here, the forcing cone angle of the barrel is also close to 3° , therefore it can be inferred that the leading angle of the band and the forcing cone angle of the barrel should be equal to achieve maximum contact force.

4. CONCLUSIONS

In this work, FE simulations of ramming operation are performed in new as well as worn-out barrel of a 155 mm×52 caliber gun system by employing the rate-dependent Johnson-Cook model. In addition, the effect of α on F_c is analysed. The performance of FE simulations is benchmarked by comparing simulated engraving marks on the band with experimental data. The important conclusions from the present study are summarized below:

For $V_r = 5$ m/s, the ramming process gets over in ~ 2.53 ms, the projectile travels a distance of ~ 4.5 mm inside the barrel before contact between band and barrel is established, and the maximum projectile displacement at the end of the ramming process is around 9 mm.

The contact force drops gradually with an increase in barrel wear up to 0.5 %, whereas it reduces rapidly for further increases in barrel wear.

The initial ramming velocity required to avoid the fallback of the projectile in a worn barrel increases marginally as the percentage wear increases up to 1 % but it increases exponentially for further increase in barrel wear.

The inclination angle of the leading surface of the band affects the extent of plastic deformation in the band. As this angle approaches the forcing cone angle of the barrel (here it is 3°), maximum contact force is generated, which is a desired condition for a successful ramming operation.

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Contribution in the current study: Visualization, project administration, writing-review & editing.

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